

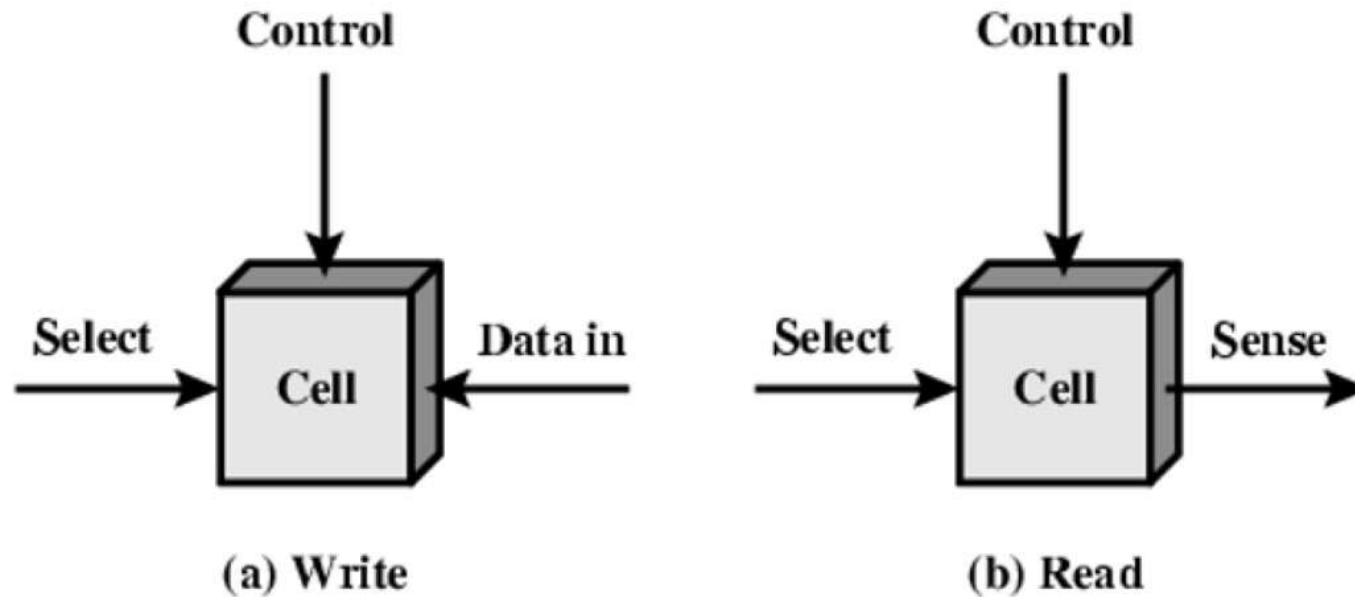
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Computer Organization
and Architecture
7th Edition

Chapter 5
Internal Memory

Big Picture Motivation

- Processor can compute only as fast as memory
 - A 3Ghz processor can execute an “add” operation in 0.33ns
 - Today’s “Main memory” latency is more than 33ns
 - Naïve implementation: loads/stores can be 100x slower than other operations
- Unobtainable goal:
 - Memory that operates at processor speeds
 - Memory as large as needed for all running programs
 - Memory that is cost effective
- Can’t achieve all of these goals at once

Memory Cell Operation



- Has two state: 0 or 1
- Can be written to set the state.
- Can be read to sense the state.

Semiconductor Memory Types

Memory Type	Category	Erasure	Write Mechanism	Volatility
Random-access memory (RAM)	Read-write memory	Electrically, byte-level	Electrically	Volatile
Read-only memory (ROM)	Read-only memory	Not possible	Masks	Nonvolatile
Programmable ROM (PROM)			Electrically	
Erasable PROM (EPROM)	Read-mostly memory	UV light, chip-level		
Electrically Erasable PROM (EEPROM)		Electrically, byte-level		
Flash memory		Electrically, block-level		

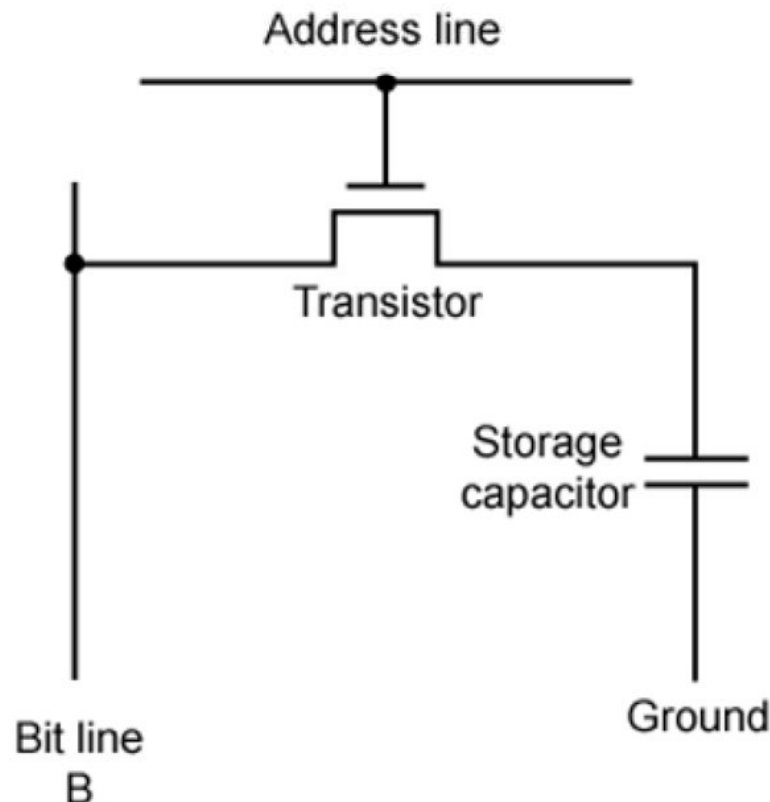
RAM

- Write / read data easily and rapidly by using electrical signal.
- Volatile. Temporary storage.
- Two types of RAM:
 - DRAM (Dynamic RAM)
 - SRAM (Static RAM)

Types of RAM (DRAM)

- **Dynamic RAM (DRAM)**

- store data as charge on **capacitors**.
- Capacitor has **discharge** tendency.
- Periodic **charge refreshing**.



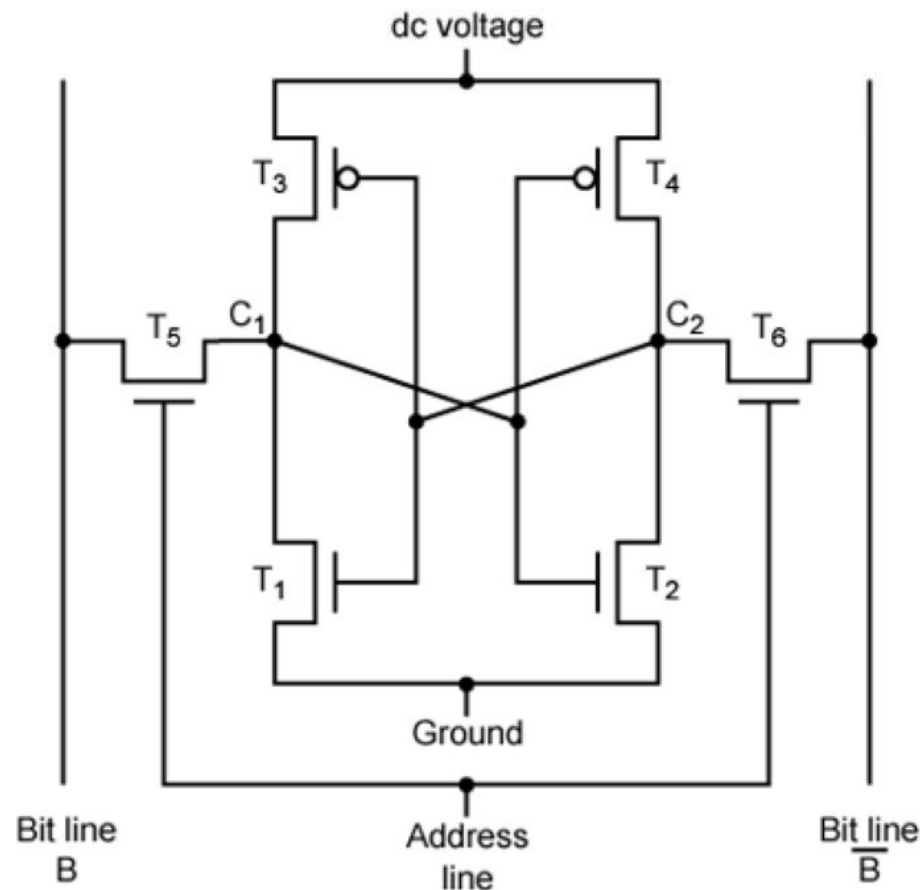
DRAM Operation

- Address line active when bit read or written
 - Transistor switch closed (current flows)
- Write
 - Voltage to bit line
 - High for 1 low for 0
 - Then signal address line
 - Transfers charge to capacitor
- Read
 - Address line selected
 - transistor turns on
 - Charge from capacitor fed via bit line to sense amplifier
 - Compares with reference value to determine 0 or 1
 - Capacitor charge must be restored

Types of RAM (SRAM)

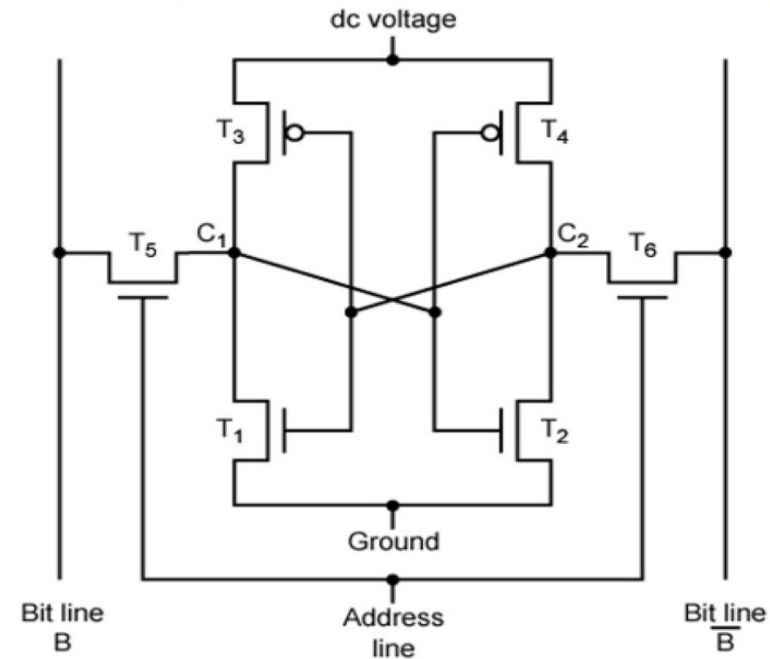
- **Static RAM (SRAM)**

- Binary value **stored** using **traditional flip-flop logic-gate** configurations.
- **Holds** data as long as **power is supplied**.



Static RAM Operation

- Transistor arrangement gives stable logic state
- State 1
 - C_1 high, C_2 low
 - T_1 T_4 off, T_2 T_3 on
- State 0
 - C_2 high, C_1 low
 - T_2 T_3 off, T_1 T_4 on
- Address line transistors T_5 T_6 is switch
- Write – apply value to B & compliment to $\overline{B}$
- Read – value is on line B



SRAM v DRAM

- Both volatile
 - Power needed to preserve data
- Dynamic RAM
 - Simpler to build, smaller
 - More dense (chip size)
 - Less expensive
 - Needs refresh**
- Static RAM
 - Faster
- Dynamic RAM used for **memory** and Static RAM used for **cache**.

Read Only Memory (ROM)

- Permanent storage
 - Nonvolatile (No power source required to maintain the bit value.)
- Used to save:
 - Library subroutines for frequently wanted function.
 - Systems programs (BIOS)
 - Function tables

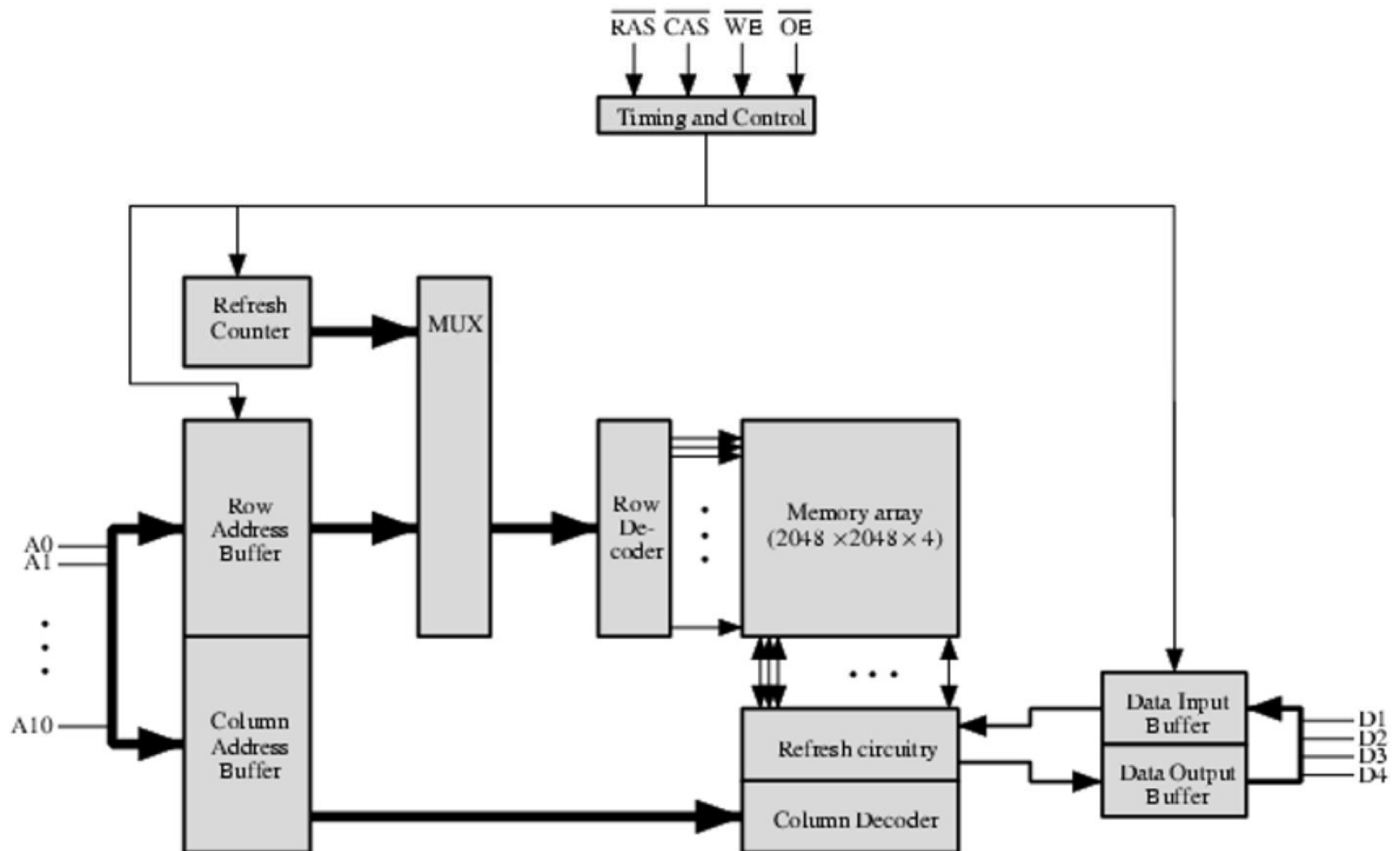
Types of ROM

- ROM
 - All data written during manufacture and non-erasable.
 - Large production** and **No room for error.**
- PROM
 - Can be **written once** only and non-erasable.
 - Needs special equipment to program
 - Used for special requirement and small numbers production.

Types of ROM

- Read “mostly” (mostly used for read operation)
 - Erasable Programmable (EPROM)
 - Erased by UV before write. (all data erased)
 - Take long time to erase.
 - Electrically Erasable (EEPROM)
 - Write any time without erase the previous content.
 - Takes much longer to write than read
 - More expensive and less dense than EPROM.
 - Flash memory
 - Fast to be programmed.
 - Erase whole memory electrically and faster than EPROM.
 - No byte erase but higher density compared to EEPROM.

Typical 16 Mb DRAM (4M x 4)



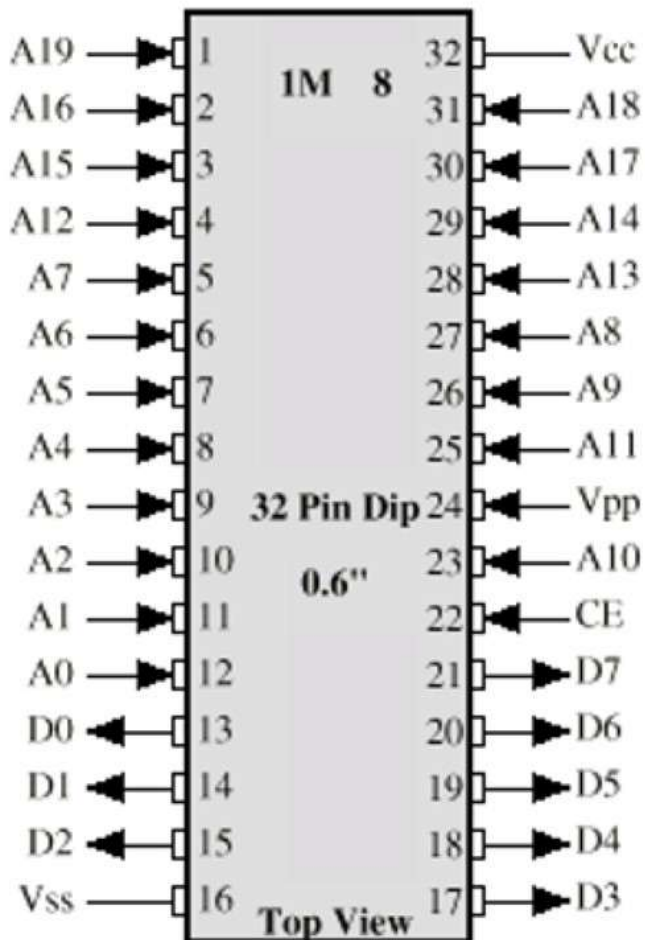
Organisation in detail

- A 16Mbit chip can be organised as 1M of 16 bit words
- A 16Mbit chip can be organised as a 2048 x 2048 x 4bit array
 - Reduces number of address pins
 - Multiplex row address and column address
 - 11 pins to address ($2^{11}=2048$)
 - Adding one more pin doubles range of values so x4 capacity

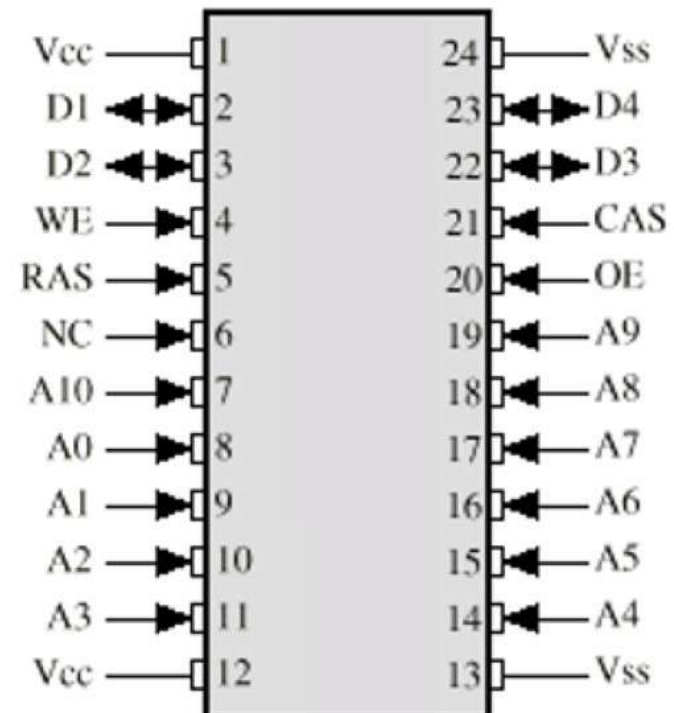
Refreshing

- Refresh circuit included on chip
- Disable chip
- Count through rows
- Read & Write back
- Takes time
- Slows down apparent performance

Packaging



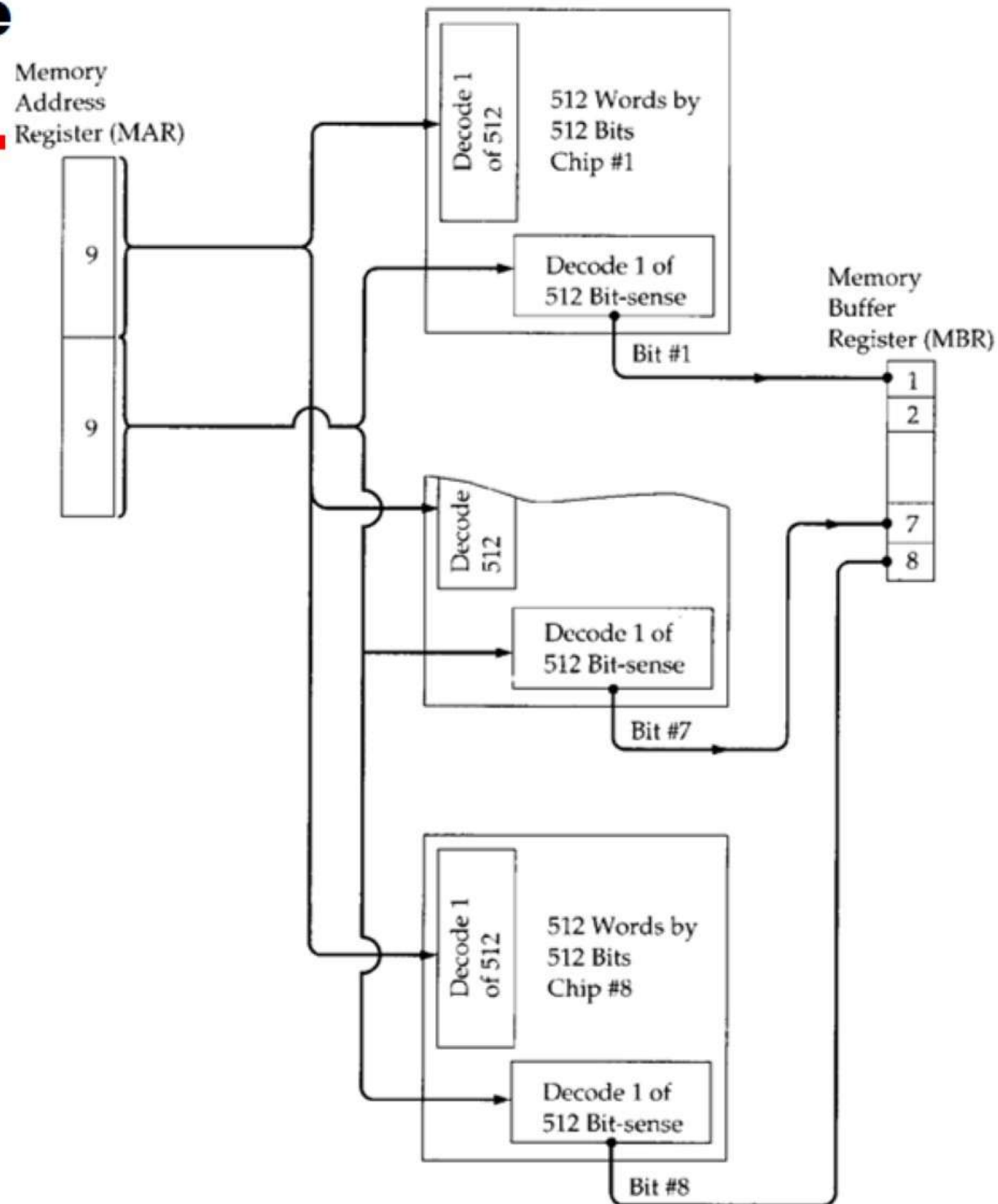
(a) 8 Mbit EPROM



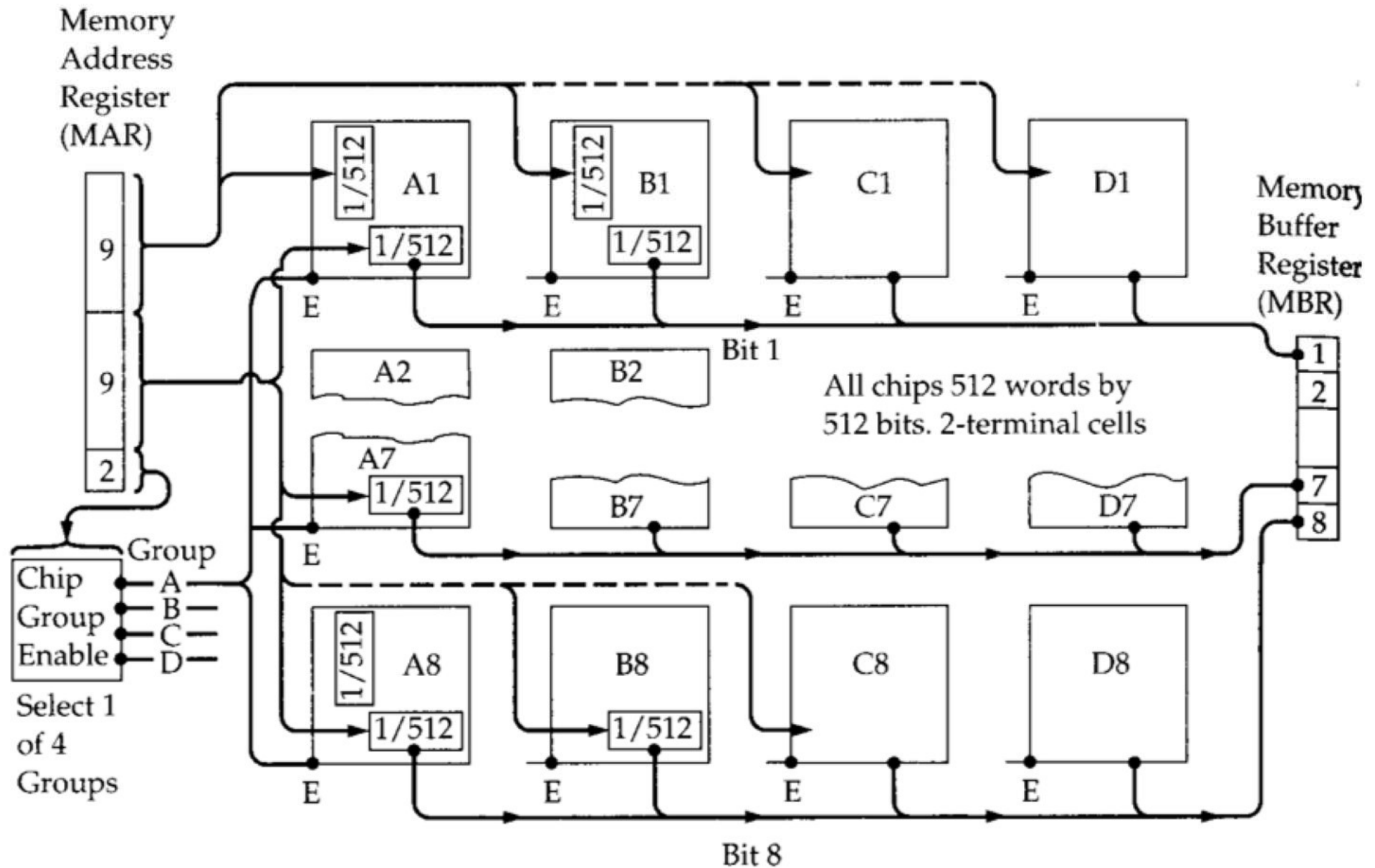
(b) 16 Mbit DRAM

256kByte Module Organisation

- Memory module
 - Combine **Eight** 1 bit RAM to get a 8 bit data.



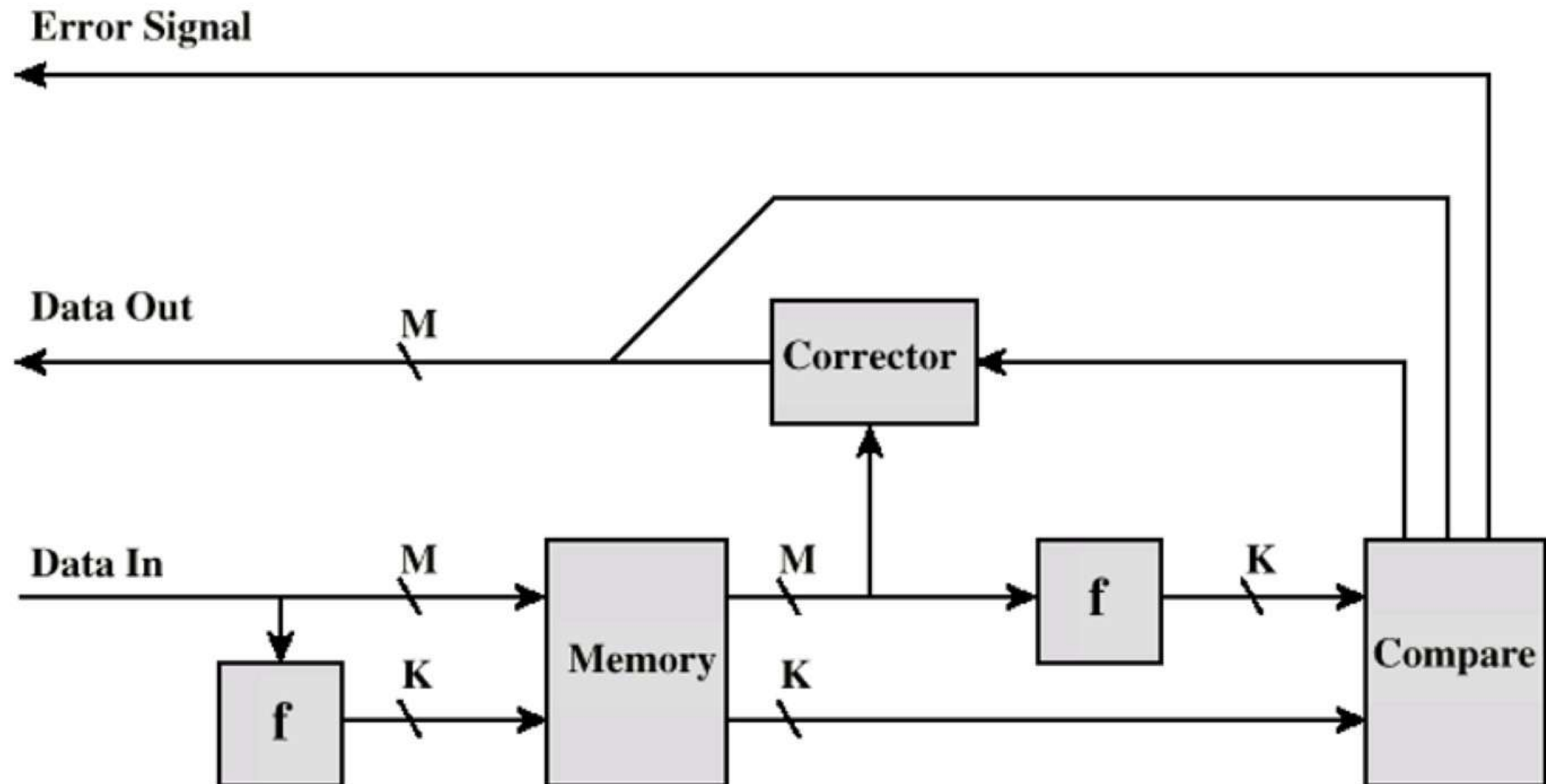
1MByte Module Organisation



Error Correction

- Hard Failure
 - Permanent defect
- Soft Error
 - Random, non-destructive
 - No permanent damage to memory
- Detected using Hamming error correcting code

Error Correcting Code Function



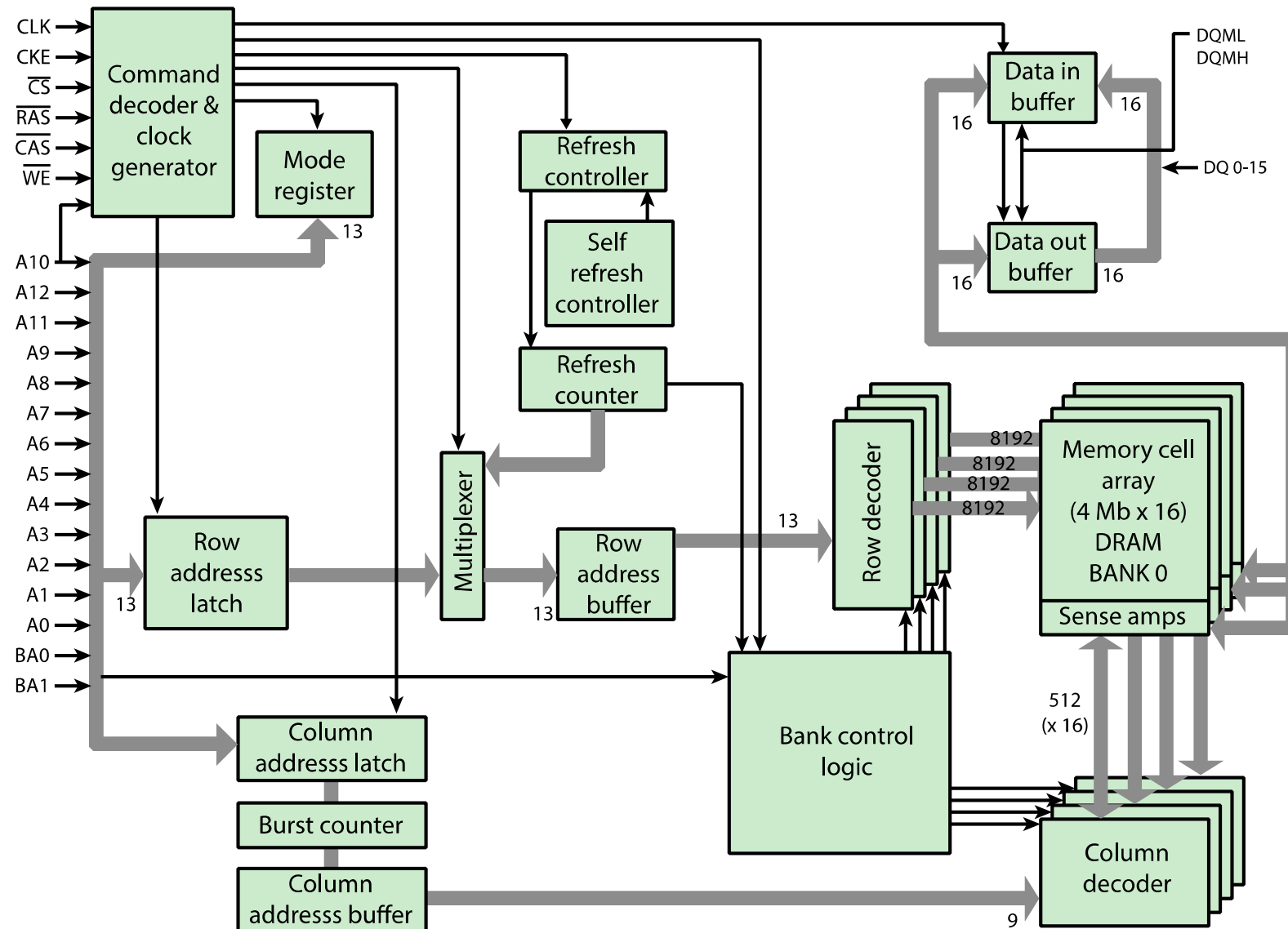
Advanced DRAM Organization

- Memory performance affect the computer performance.
- SRAM is included between the DRAM and processor to improve the speed.
- Recent types of DRAM:
 - SDRAM
 - DDR-RAM
 - RDRAM
 - CDRAM

Synchronous DRAM (SDRAM)

- Access is synchronized with an external clock
- Address is presented to RAM
- RAM finds data (CPU waits in conventional DRAM)
- Since SDRAM moves data in time with system clock, CPU knows when data will be ready
- CPU does not have to wait, it can do something else
- Burst mode allows SDRAM to set up stream of data and fire it out in block
- DDR-SDRAM sends data twice per clock cycle (leading & trailing edge)

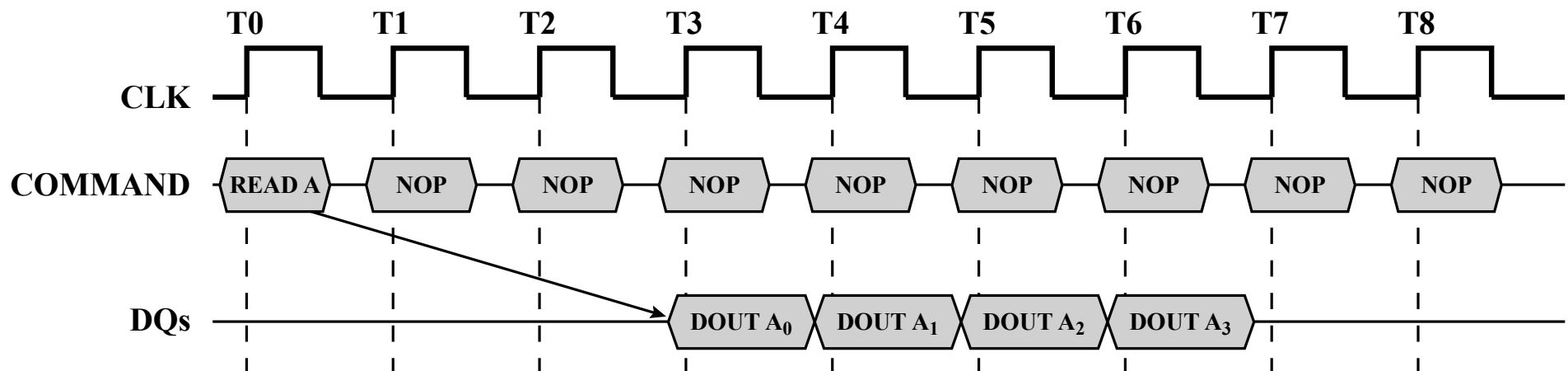
256-Mb Synchronous Dynamic RAM (SDRAM)



SDRAM Pin Assignments

A0 to A13	Address inputs
BA0, BA1	Bank address lines
CLK	Clock input
CKE	Clock enable
CS	Chip select
RAS	Row address strobe
CAS	Column address strobe
WE	Write enable
DQ0 to DQ7	Data input/output
DQM	Data mask

SDRAM Read Timing (burst length = 4, $\overline{\text{CAS}}$ latency = 2)



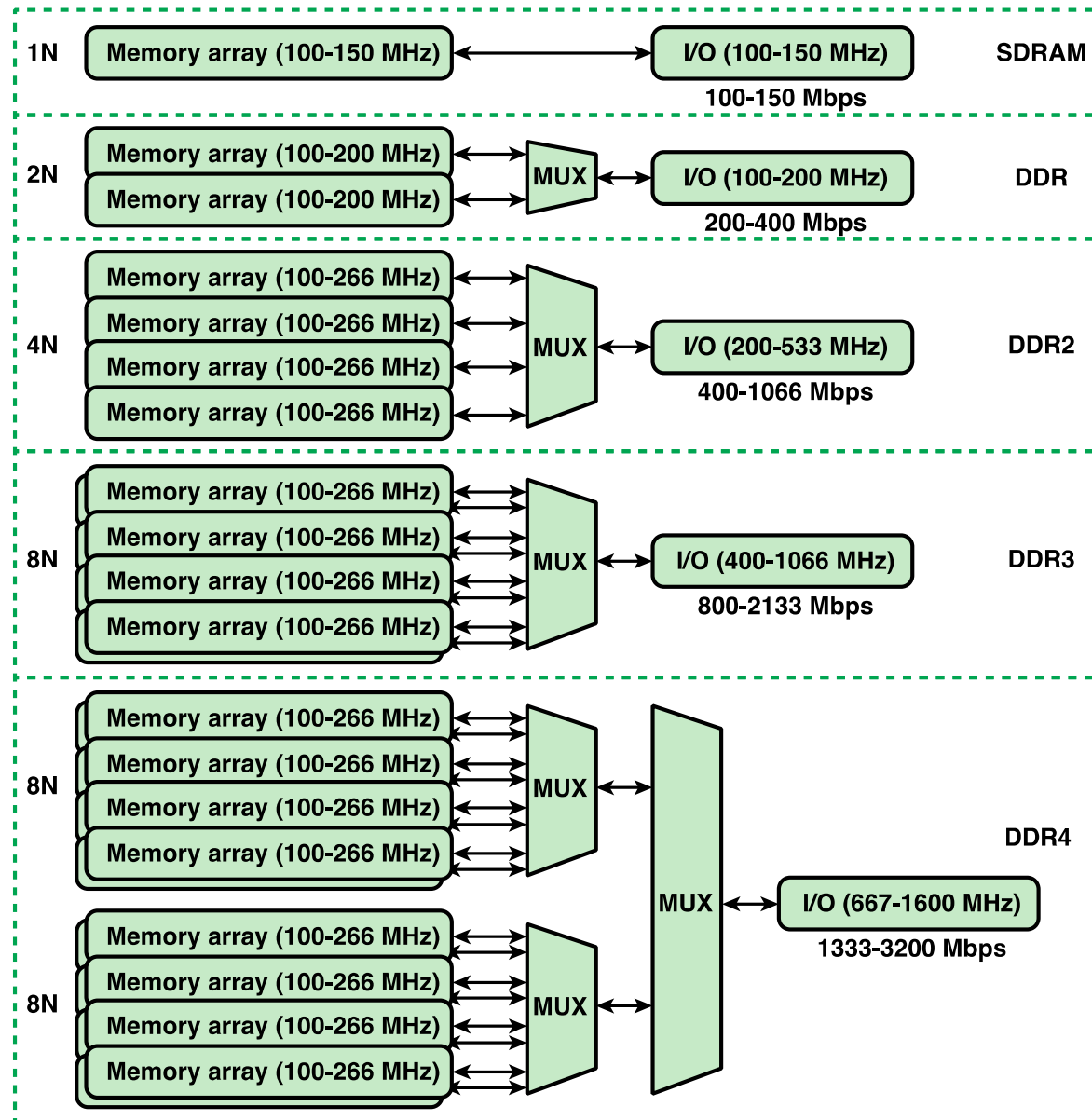
Double Data Rate SDRAM (DDR SDRAM)

- Developed by the JEDEC Solid State Technology Association (Electronic Industries Alliance's semiconductor-engineering-standardization body)
- Numerous companies make DDR chips, which are widely used in desktop computers and servers
- DDR achieves higher data rates in three ways:
 - First, the data transfer is synchronized to both the rising and falling edge of the clock, rather than just the rising edge
 - Second, DDR uses higher clock rate on the bus to increase the transfer rate
 - Third, a buffering scheme is used

DDR Characteristics

	DDR1	DDR2	DDR3	DDR4
Prefetch buffer (bits)	2	4	8	8
Voltage level (V)	2.5	1.8	1.5	1.2
Front side bus data rates (Mbps)	200—400	400—1066	800—2133	2133—4266

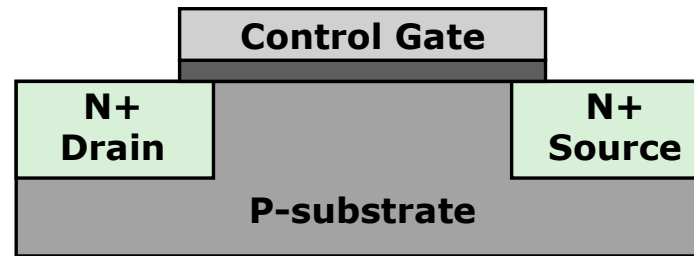
DDR Generations



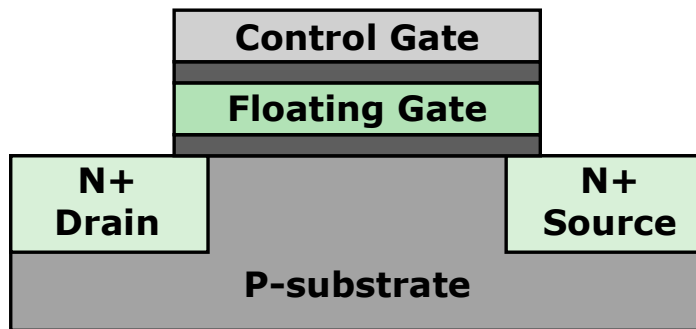
Flash Memory

- Used both for internal memory and external memory applications
- First introduced in the mid-1980's
- Is intermediate between EPROM and EEPROM in both cost and functionality
- Uses an electrical erasing technology like EEPROM
- It is possible to erase just blocks of memory rather than an entire chip
- Gets its name because the microchip is organized so that a section of memory cells are erased in a single action
- Does not provide byte-level erasure
- Uses only one transistor per bit so it achieves the high density of EPROM

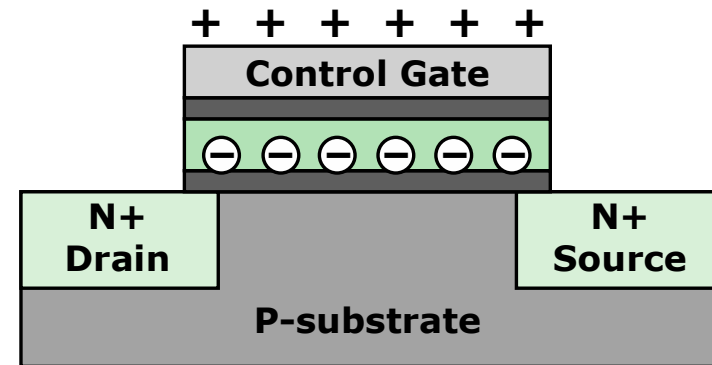
Flash Memory Operation



(a) Transistor structure



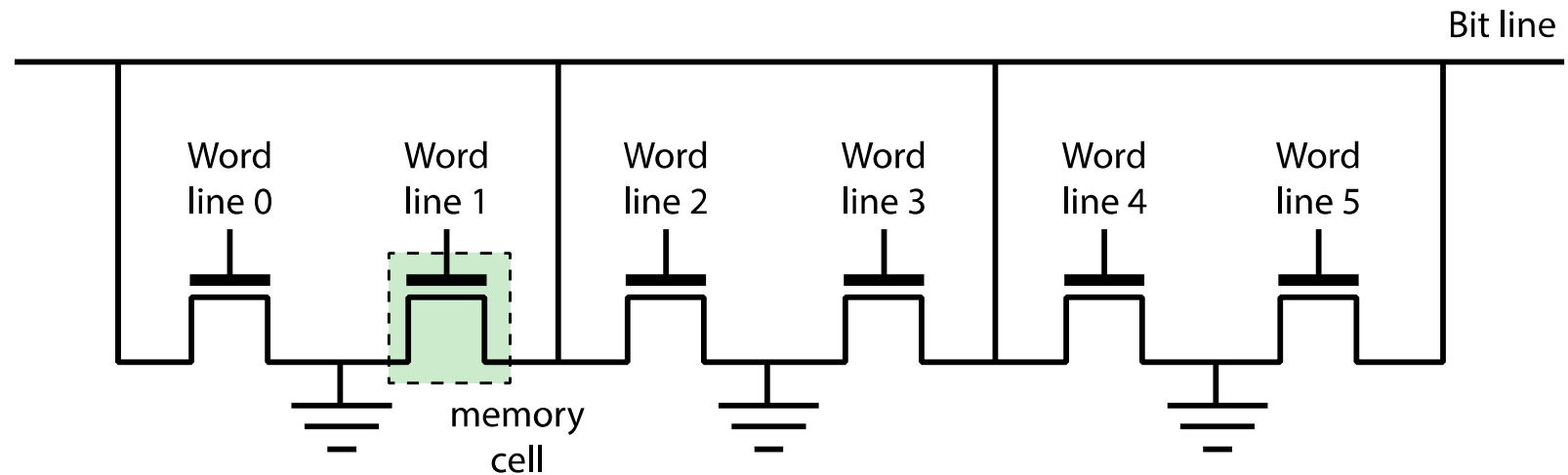
(b) Flash memory cell in one state



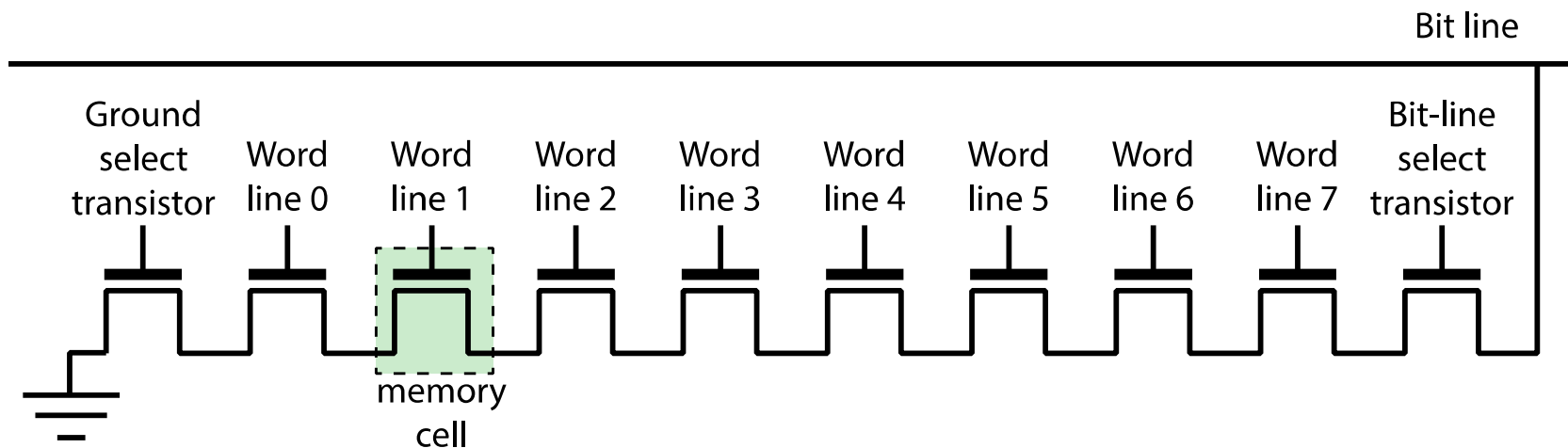
(c) Flash memory cell in zero state

Figure 6.18

Flash Memory Structures



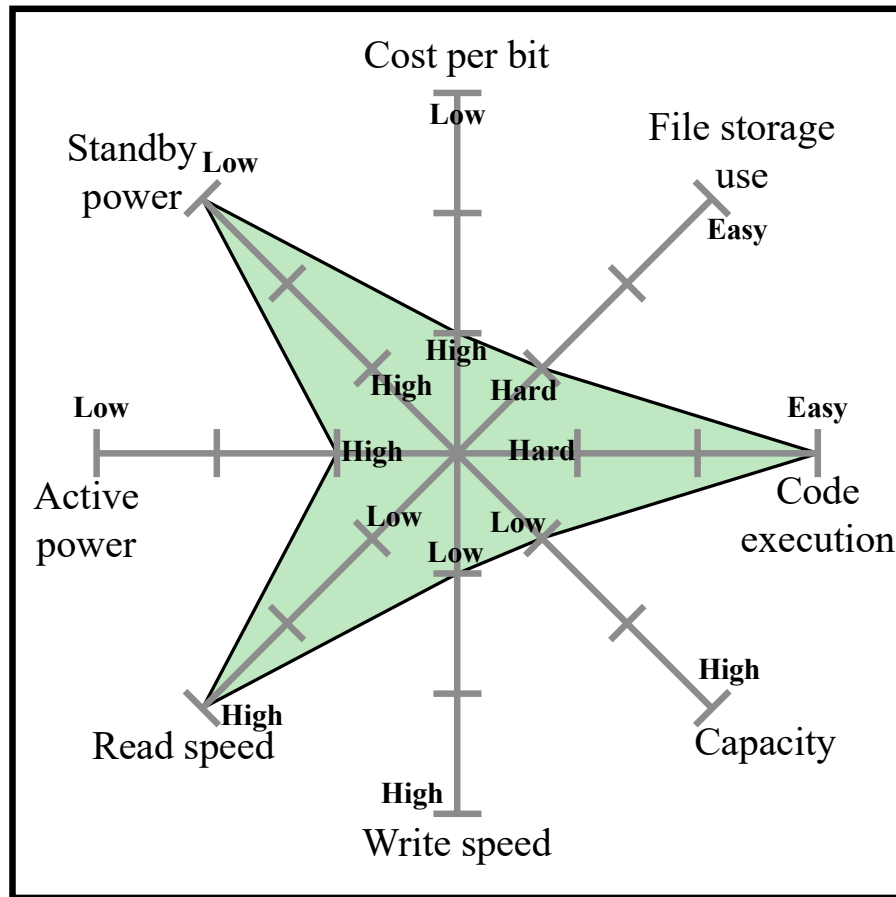
(a) NOR flash structure



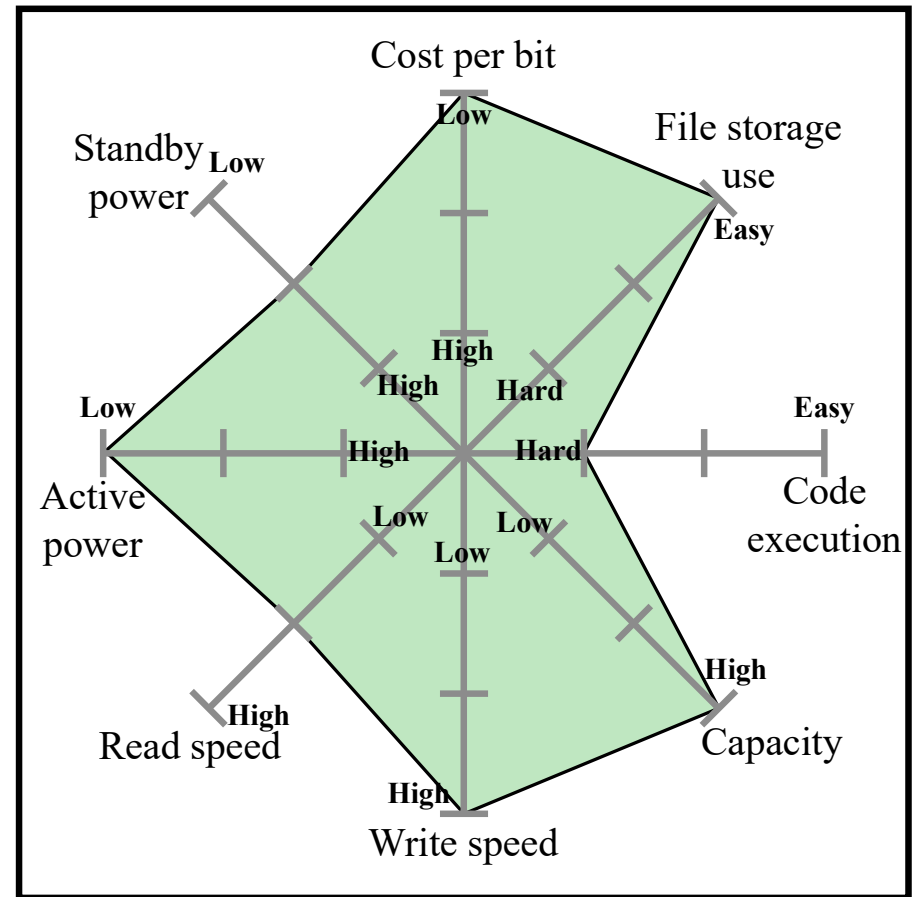
(b) NAND flash structure

Figure 6.19

Kiviat Graphs for Flash Memory



(a) NOR



(b) NAND

Figure 6.20

Nonvolatile RAM within the Memory Hierarchy

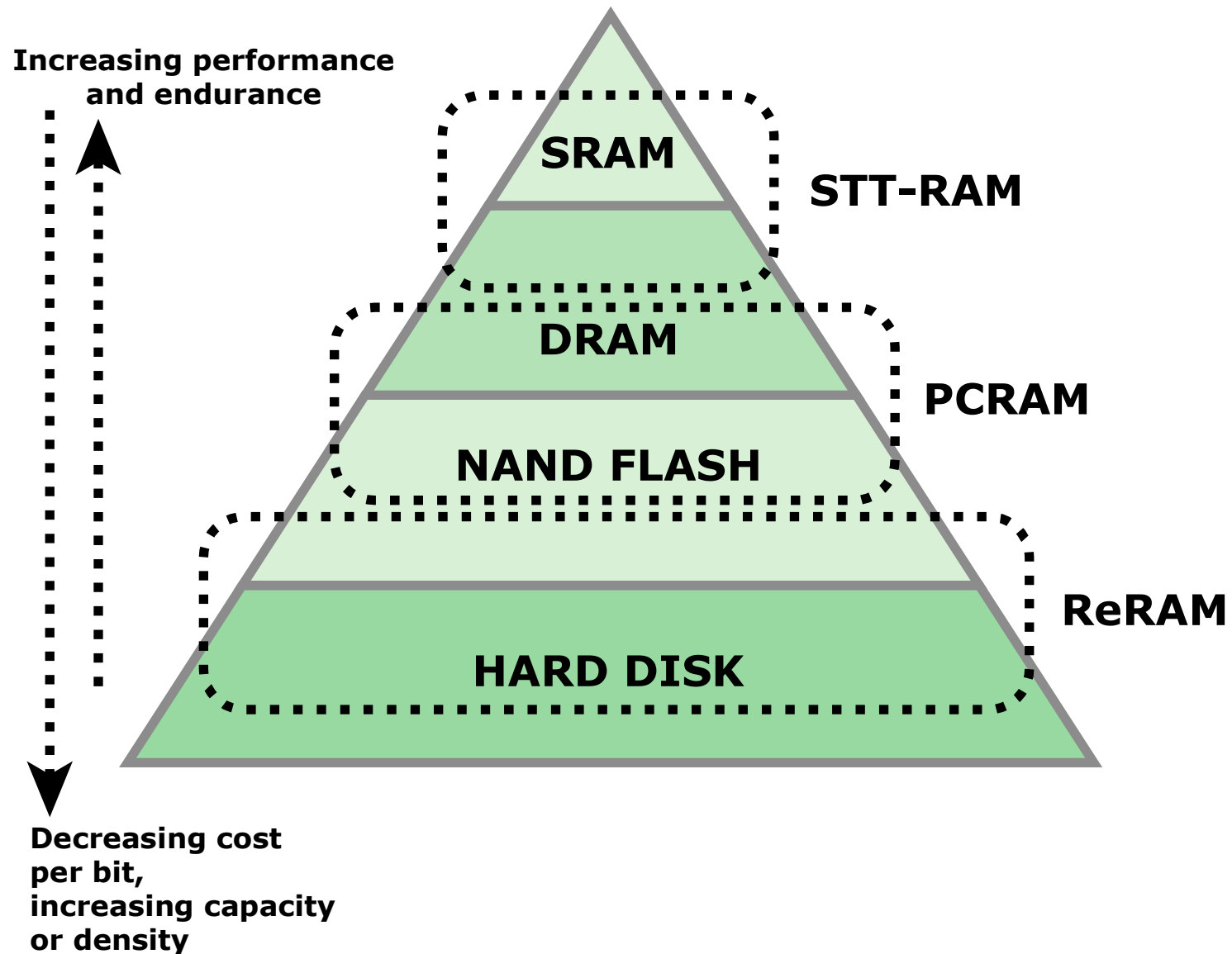
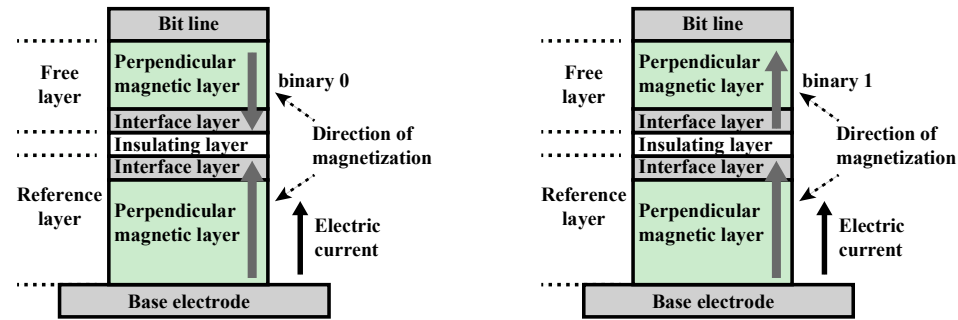
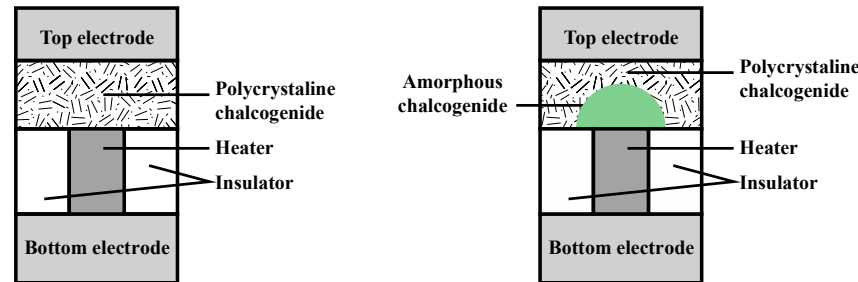


Figure 6.21

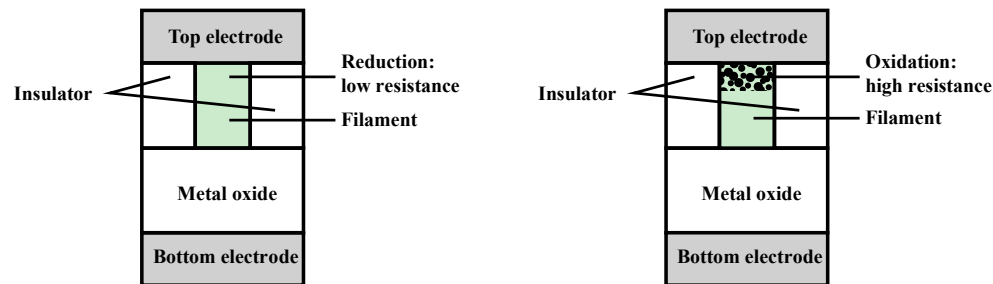
Nonvolatile RAM Technologies



(a) STT-RAM



(b) PCRAM



(c) ReRAM